AFFORDABLE RENEWABLE ENERGY TRADING

Milestone: Implementation in NoSQL

Group 12

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Percentage of Effort Contributed by Student1: 50%

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Connection to NoSQL Database – NEO4J : using cypher

EXPLORER

WEBDEV PROJECTS
- DMA-Userproject
  - __pycache__
  - database
  - .env.txt
  - analytics.py
  - app.py
  - data_load.py
  - solshareNeo4J_config...
  - tempCodeRunnerFile....
- Memories project
  - MemoryGallery
  - mernProjectEcommer...
- Spotify Clone

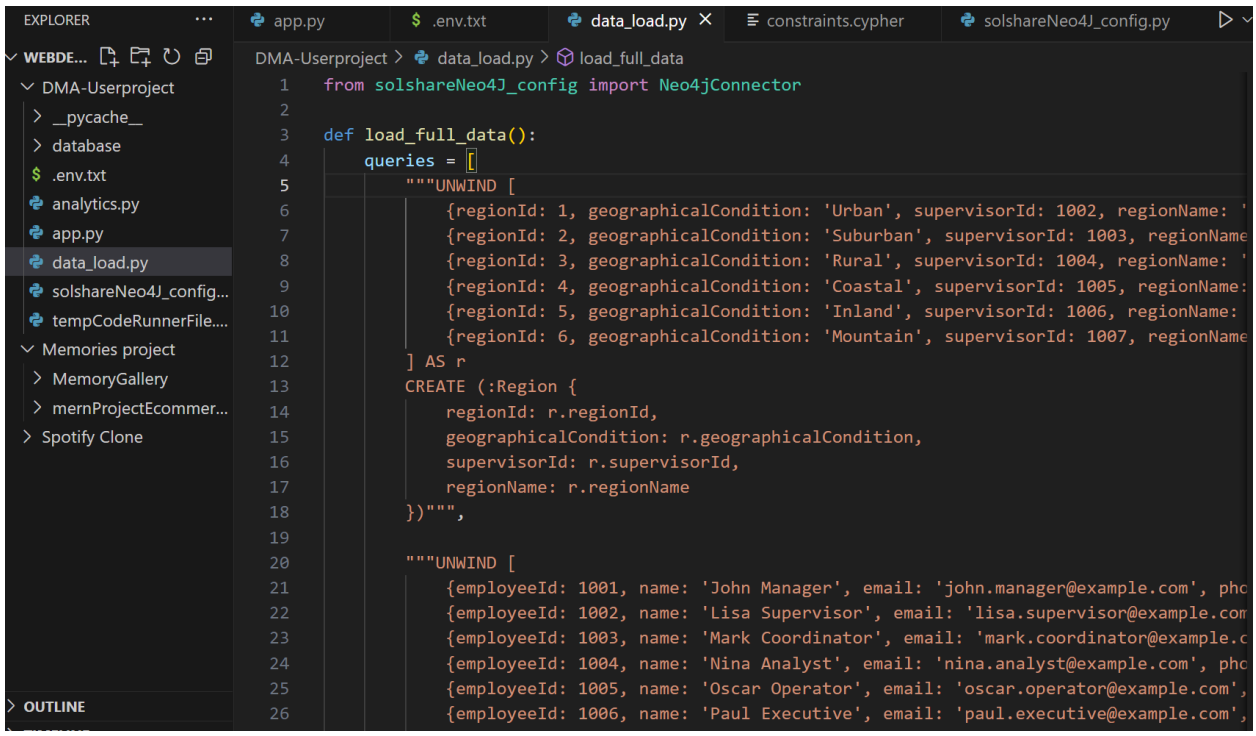
OUTLINE

app.py | .env.txt | data_load.py | constraints.cypher | solshareNeo4J_config.py

DMA-Userproject > solshareNeo4J_config.py > Neo4jConnector

```python
from neo4j import GraphDatabase
from dotenv import load_dotenv
import os

load_dotenv()

class Neo4jConnector:
    def __init__(self):
        self.uri = os.getenv("NEO4J_URI", "bolt://localhost:7687")
        self.user = os.getenv("NEO4J_USER", "neo4j")
        self.password = os.getenv("NEO4J_PASSWORD", "12345678")
        self.driver = GraphDatabase.driver(self.uri, auth=(self.user, self.password))

    def execute_query(self, query, parameters=None):
        with self.driver.session() as session:
            return session.run(query, parameters).data()

    def close(self):
        self.driver.close()

# Test connection
if __name__ == "__main__":
    conn = Neo4jConnector()
    print(conn.execute_query("RETURN 'Connection successful' AS message"))
    conn.close()
```

Loaded Database using python files separately :



SOME QUERIES IN NOSQL DATABASE – [CYPHER]

1. Find Users in a Specific Region

MATCH (u:User)-[:LOCATED_IN]->(r:Region {regionName: 'Eastern Boston'})
RETURN DISTINCT u.userId, u.name, u.email, u.phone
ORDER BY u.userId;

```
neo4j$ MATCH (u:User)-[:LOCATED_IN]→(r:Region {regionName: 'Eastern Boston'}) RETURN DISTINCT u.use…
```

Table | Text | Code

u.userId	u.name	u.email	u.phone
54322	"Brian Smith"	"brian.smith@example.com"	"617-555-0102"
54327	"Grace Taylor"	"grace.taylor@example.com"	"617-555-0107"
54334	"Nathan Harris"	"nathan.harris@example.com"	"617-555-0114"
54340	"Tina Allen"	"tina.allen@example.com"	"617-555-0120"
54345	"Epsilon Trading"	"epsilon.trading@example.com"	"617-555-0205"
54348	"Theta Services"	"theta.services@example.com"	"617-555-0208"
54355	"Omicron Holdings"	"omicron.holdings@example.com"	"617-555-0215"

2. Energy Production vs. Consumption by Region

```
MATCH (u:User)-[:LOCATED_IN]->(r:Region), (u)-[:HAS_PROFILE]->(p:EnergyProfile)
RETURN r.regionName AS Region,
    COUNT(u) AS UserCount,
    SUM(p.energyProduced) AS TotalProduction,
    SUM(p.energyConsumed) AS TotalConsumption,
    SUM(p.surplusEnergy) AS TotalSurplus
ORDER BY TotalSurplus DESC;
```

neo4j$ MATCH (u:User)-[:LOCATED_IN]→(r:Region), (u)-[:HAS_PROFILE]→(p:EnergyProfile) RETURN r.regi…

Table | Text | Code

Region	UserCount	TotalProduction	TotalConsumption	TotalSurplus
"Western Cambridge"	600	128625.0	124252.49999999985	4372.500000000005
"Eastern Boston"	525	109500.0	105464.99999999985	4035.0000000000045
"Northern Worcester"	525	106500.0	103031.25	3468.75
"Southern Salem"	450	91500.0	88200.0	3300.0
"Highland Newton"	450	88125.0	84881.25	3243.75
"Central Lowell"	450	92625.0	89400.0	3225.0

3. Identify Top Energy Producers

MATCH (u:User)-[:HAS_PROFILE]->(p:EnergyProfile)
RETURN u.name AS UserName, p.energyProduced AS EnergyProduce
d
ORDER BY p.energyProduced DESC LIMIT 10;

```
neo4j$ MATCH (u:User)-[:HAS_PROFILE]→(p:EnergyProfile) RETURN u.name AS UserName, p.energyProduced …
```

Table | Text | Code

UserName	EnergyProduced
"Zeta Logistics"	245.0
"Nu Energy"	245.0
"Upsilon Media"	245.0
"Zeta Logistics"	245.0
"Zeta Logistics"	245.0
"Zeta Logistics"	245.0
"Zeta Logistics"	245.0
"Nu Energy"	245.0
"Nu Energy"	245.0
"Nu Energy"	245.0

4. Energy Grid Performance Analysis.

```
MATCH (u:User)-[:MEMBER_OF]->(g:Grid), (u)-[:HAS_PROFILE]->(p:EnergyProfile)
WITH g, COUNT(u) AS MemberCount,
    SUM(p.energyProduced) AS TotalProduction,
    SUM(p.energyConsumed) AS TotalConsumption
RETURN g.connectionId AS GridID,
      MemberCount,
      TotalProduction,
      TotalConsumption,
      TotalProduction - TotalConsumption AS NetOutput
ORDER BY NetOutput DESC;
```

neo4j$ MATCH (u:User)-[:MEMBER_OF]→(g:Grid), (u)-[:HAS_PROFILE]→(p:EnergyProfile) WITH g, COUNT(u)…

Table | Text | Code

GridID	MemberCount	TotalProduction	TotalConsumption	NetOutput
8005	165	33825.0	32175.0	1650.0
8006	165	28875.0	27225.0	1650.0
8004	165	29975.0	28352.5	1622.5
8001	165	26400.0	24802.25000000002	1597.7499999999818
8002	165	32175.0	30577.25000000002	1597.7499999999818
8003	165	29700.0	28105.0	1595.0
8005	150	30750.0	29250.0	1500.0
8006	150	26250.0	24750.0	1500.0
8004	150	27250.0	25775.0	1475.0
8001	150	24000.0	22547.50000000002	1452.4999999999818

5. Geographical Energy Production Efficiency

```
MATCH (r:Region)<-[:LOCATED_IN]-(u:User)-[:HAS_PROFILE]->(p:EnergyProfile)
WITH r, u, p
RETURN r.geographicalCondition AS GeographicalCondition,
    AVG(p.energyProduced) AS AvgProduction,
    AVG(u.renewableCapacity) AS AvgCapacity,
    AVG(p.energyProduced)/AVG(u.renewableCapacity) AS EfficiencyRatio
ORDER BY EfficiencyRatio DESC;
```

neo4j$ MATCH (r:Region)←[:LOCATED_IN]-(u:User)-[:HAS_PROFILE]→(p:EnergyProfile) WITH r, u, p RETUR…

Table | Text | Code

GeographicalCondition	AvgProduction	AvgCapacity	EfficiencyRatio
"Inland"	205.83333333333334	29.250000000000018	7.037037037037033
"Coastal"	203.33333333333323	29.291666666666664	6.941678520625886
"Rural"	202.85714285714272	29.999999999999993	6.761904761904759
"Mountain"	195.83333333333343	30.174999999999976	6.489919911626631
"Suburban"	214.37500000000003	34.18749999999999	6.270566727605121
"Urban"	208.57142857142838	33.35714285714287	6.2526766595289

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